

# Anwin Sen Padma

## Backend Engineer — Distributed Systems & AI-Enabled Platforms

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### PROFILE

Engineer with 15 years building and shipping high-performance APIs and distributed systems, from founding the full product stack of a real-time drone intelligence platform to technically leading the backend behind a global 5G enterprise product. Go, Python (FastAPI), PostgreSQL, Kafka, and Kubernetes are day-to-day tools, covering the full range: API design, business logic, deployment, observability, infrastructure. More recently been working with AI agentic tooling: authored Kiro steering files across EP5G's Go backend repos (covering repo architecture, coding standards, and domain context) and integrated MCP servers into EP5G backend APIs to connect with AI AgentCore. Writes clean, well-tested code and takes ownership from first commit through to production.

### CORE SKILLS

|               |                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages     | Go (primary), Python (FastAPI, Flask), TypeScript, C++                                                                                                                      |
| Data          | PostgreSQL (high-throughput transactional systems, schema design, multi-tenancy), MongoDB, DynamoDB, Redis. Distributed data modelling at scale.                            |
| Messaging     | Kafka (async orchestration, event-driven architecture, real-time pipelines), gRPC, RabbitMQ, WebSockets                                                                     |
| Platform      | Kubernetes, Helm, Docker, Terraform, AWS (EKS, Lambda, RDS, S3), GCP, Azure, Envoy, GitLab CI/CD. Full ownership from code to deployment.                                   |
| Observability | Prometheus, Grafana, ELK Stack, distributed tracing. SLI/SLO design, alerting, tail latency optimisation.                                                                   |
| Architecture  | Microservices, Event-Driven, REST and gRPC API design, multi-tenancy, authentication and authorisation, developer experience, resilience engineering, cloud-native patterns |
| Leadership    | Cross-functional alignment, architecture review, staff mentoring, roadmap shaping                                                                                           |
| Frontend      | React (TypeScript), configuration UI for EP5G.                                                                                                                              |
| AI Tooling    | Kiro AI agent (steering files, repo context, coding standards), MCP server integration, AI AgentCore, agentic workflow design                                               |

### KEY ACHIEVEMENTS

- Hands-on with AI agentic coding at Ericsson using Kiro: authored steering files across multiple EP5G Go backend repos covering architecture, service boundaries, and coding standards, giving the agent enough context to actually help rather than produce generic output. Also integrated MCP servers into EP5G backend APIs to connect live services with AI AgentCore.
- Founding engineer at Asteria Aerospace. Designed and shipped the full product stack: backend APIs in Python and Go, real-time drone fleet management, AI-powered video classification, Kafka/WebSocket event pipelines, through to Kubernetes deployment and production ops.
- Technical lead on EP5G's Go backend: multi-tenant PostgreSQL configuration APIs, high-throughput Kafka ingest pipelines across thousands of distributed nodes, and the full delivery stack: Helm, Terraform, Prometheus, Envoy.

- ▶ Built EP5G's observability platform from scratch: ELK, Prometheus, Grafana, SLI/SLO definitions, and internal Go SDKs so teams had a consistent way to plug into shared services. MTTR came down noticeably once teams had proper visibility into what was actually failing.
- ▶ At Emperia, shipped cloud-native FastAPI/Python backend services on serverless AWS (Lambda, API Gateway, Terraform) with JWT/Auth0 auth and near-complete Pytest coverage. Worked directly with product and frontend throughout, so the feedback loop was short.
- ▶ Worked with Product leadership on EP5G's technical roadmap, mentored engineers across backend, ML, and DevOps, and put real time into code reviews and design standards.

## PROFESSIONAL EXPERIENCE

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### Senior Software Engineer — EP5G Platform

*May 2022 – Present*

Ericsson | Camberley, UK

- ▶ Integrated MCP servers into EP5G backend APIs to connect with AI AgentCore, and rolled out Kiro AI agent steering files across Go backend repos, documenting architecture, service boundaries, and coding standards so the agent has genuine repo context across the EP5G Go backend.
- ▶ Technical lead on EP5G's Go backend: shipped product features, REST and gRPC APIs, multi-tenant PostgreSQL configuration services, and Kafka ingest pipelines handling real-time orchestration across thousands of distributed nodes. Applied idempotency, backpressure, and retry strategies to keep latency under control at scale.
- ▶ Owned the full delivery stack: Go features, Helm chart changes, Terraform infrastructure, Prometheus metrics and alerting, Envoy gateway config. Also built EP5G's observability platform from scratch: ELK, Grafana, SLI/SLO policies, and internal Go SDKs that gave teams a common way to work with shared services.
- ▶ Drove CI/CD automation across GitLab, Kubernetes, Docker, and Salt, and built internal Go SDKs and developer tooling to give squads reusable foundations rather than wiring the same things up from scratch each time.
- ▶ Worked with Product leadership on platform architecture direction, mentored engineers across all levels, and put consistent effort into code reviews and design standards.
- ▶ Contributed to EP5G's Network Management Portal (NMP), a React/TypeScript web app, building configuration pages for multi-site setup and a Console/Terminal interface for executing CLI commands against network controllers.

### Lead Backend Developer

*Nov 2021 – May 2022*

Emperia Ltd | London, UK

- ▶ Designed and shipped cloud-native Python/FastAPI backend services with MongoDB Atlas, building customer-facing APIs powering virtual store experiences for luxury brands. Worked closely with product and customers throughout, which kept the work focused on what actually mattered.
- ▶ Architected serverless AWS infrastructure using Lambda, API Gateway, ECR, and Terraform. Auto-scaling pipelines with a noticeably lower hosting bill than running always-on servers.
- ▶ Built secure REST APIs with JWT/Auth0 authentication; implemented GitHub Actions CI/CD achieving near-complete automated test coverage with Pytest.
- ▶ Worked directly with product and frontend teams, which meant priorities stayed practical and decisions got made faster.

### Staff Software Engineer — Founding Team

*May 2013 – Nov 2021*

Asteria Aerospace | Bangalore, India

- ▶ Designed and shipped the full product as a founding engineer at Asteria Aerospace: backend APIs and business logic in Python and Go, Drone Mission Control application (Qt/C++), centralised fleet management features, AI-powered aerial video classification, and async event pipelines on Kafka and WebSockets backed by PostgreSQL and MongoDB.
- ▶ Built async event pipelines on Kafka and WebSockets as part of the product backend, decoupling drone fleet services, enabling real-time data flow across the platform, and handling high-frequency sensor streams with sub-second end-to-end latency as the fleet scaled.
- ▶ Led integration of Azure Custom Vision-powered vision analytics into live video feeds, partnering with the ML team to define the data pipeline architecture and deploy real-time inference at the edge.

- ▶ Worked closely across backend, ML, and DevOps teams to define API contracts and service boundaries.
- ▶ Set up engineering practices from day one: TDD, CI/CD with Jenkins, code review standards, agile. Small team at the start, so getting this right early meant the habits stuck as the team grew across backend, ML, and DevOps.
- ▶ Mentored backend, ML, and DevOps engineers and handled architecture questions across the team.

### **Software Developer**

*Dec 2011 – Apr 2013*

VCreate Logic | Bangalore, India

- ▶ Delivered a GIS sensor visualisation platform for SolData (France), rendering real-time data from thousands of sensors across Toulon's underground tunnel infrastructure. Built custom Qt/C++ graphical items for high-performance geospatial rendering across a large live sensor network.

### **Software Professional Trainee**

*Jul 2011 – Nov 2011*

One View Systems | Thiruvananthapuram, India

- ▶ Designed and built a Linux-based video surveillance system handling real-time multi-stream IP camera feeds. Early experience with low-latency, high-throughput constraints that have come up in most roles since.

## **EDUCATION**

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**B.Tech – Computer Science & Engineering** | Model Engineering College, CUSAT, India | 2007–2011